LLM-as-a-Judge Reliability

Phase 4 Combined Presentation

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Category 2: Bias Quantification & Fairness Analysis

Paper 3:

AN EMPIRICAL ANALYSIS OF UNCERTAINTY IN LARGE LANGUAGE MODEL EVALUATIONS

Category Alignment & Positioning

- **Taxonomy Alignment:** Category 2 (Bias Quantification & Fairness Analysis)
- **Beyond Static Prejudices:** While prior literature focuses heavily on static evaluator biases (e.g., length, position, and brand preference), this work explores **dynamic stability**.
- **Internal States:** Shifting focus from verbalized outputs to internal logit probability distributions of the evaluator.
- **Systemic Failure:** Demonstrating how hidden evaluator uncertainty leads to fragile, non-reproducible model rankings.

Paper Overview & Core Goals

- **The Core Problem:** Automated LLM judges are widely used to replace human raters, yet their internal evaluation confidence remains unquantified and volatile.
- **The Dual-Uncertainty Paradigm:** Real-world evaluations depend on a complex interaction between the judge's certainty and the candidate's generation certainty.
- **Three-Step Research Agenda:**
  1. **Existence:** Measuring raw logit probabilities to prove uncertainty exists across major model families.
  2. **Mitigation:** Testing prompt formatting styles to stabilize decision boundaries.
  3. **Utilization:** Training an uncertainty-aware judge to spot hallucinations in out-of-distribution (OOD) scenarios.

The Dual-Uncertainty Interaction

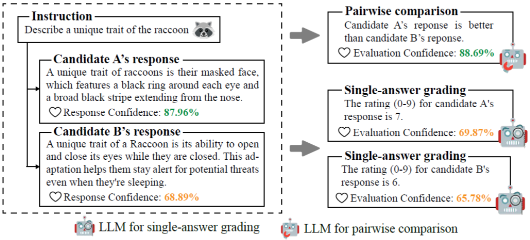

Figure 1: An example of uncertainty (i.e., model confidence) in model-based LLM evaluation. The evaluation process is influenced by the uncertainty of both the evaluator and the candidate model.

Figure 1: Visualizing the interplay of Candidate Response Confidence and Evaluator Judge Confidence.

Part 1: Quantifying Uncertainty (Existence)

- **Logit-Based Probabilities:** Instead of prompting judges to verbalize confidence (which suffers from poor numeric calibration), this work extracts the raw logits directly at the last layer of generation.
- **Evaluation Confidence:** The precise probability of the specific token representing the final decision (e.g., the score "7" in absolute grading, or "1", "2", "Tie" in pairwise comparison).
- **Response Confidence:** Calculated as the mathematical average of the generation probabilities of all tokens in the candidate's output.
- **Zero-Cost Metadata:** Extracted directly during inference, bypassing the need for computationally expensive multi-sample consistency checks.

Part 2: Mitigating Uncertainty (Mitigation)

- **Format-Induced Doubt:** Default templates prompt judges to output the score first. This forces an immediate decision under high attention entropy, dispersing the probability mass.
- **Chain-of-Thought (CoT) Calibration:** Forcing the judge to generate reasoning before the verdict token acts as a natural logit stabilizer, concentrating probability mass on the final score token.
- **Self-Generated References:** Instructing the judge to write its own reference answer first provides a solid comparative anchor, reducing rating volatility.
- **Post-Training Stabilization:** Specialized evaluator models (e.g., Prometheus2) trained in CoT formats display remarkably high confidence, but suffer from sensitivity to distribution shifts.

Part 3: Utilizing Uncertainty (ConfiLM)

- **The Ingestion Concept:** Candidate models express lower token probabilities when attempting to hallucinate facts. By feeding this uncertainty into the judge, the judge can detect errors.
- **The Verbalization Bridge:** LLMs struggle with continuous numerical float values in prompts. The authors map candidate logits into discrete natural language statements (e.g., "Highly confident", "Neutral", "Clearly doubtful").
- **ConfiLM Fine-Tuning:** Fine-tuning Llama-3-8B-Instruct on Alpaca data augmented with these verbalized confidence statements.
- **The Olympic 2024 Dataset:** A manually curated, 220-instance out-of-distribution (OOD) testing suite annotated by PhD-level experts (97.27% agreement) to evaluate real-time factual robustness.

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- **Confidence Buckets:** Ten evenly-spaced ranges from [0.0, 0.1) $\rightarrow$ *"Complete doubt"* up to [0.9, 1.0] $\rightarrow$ *"Absolute confidence"*, with *"Neutral"* at the midpoint [0.5, 0.6).
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Multi-Phase Experimental Pipeline

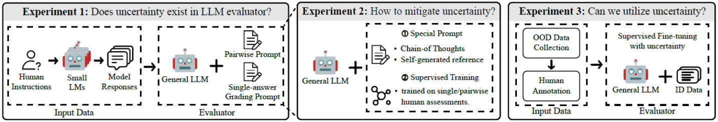

Figure 2: We conduct extensive experiments and analysis to investigate the existence, mitigation and utilization of uncertainty in model-based LLM evaluation. Uncertainty plays a key role in the evaluation process and can be leveraged to enhance the evaluator's performance in OOD scenarios.

Figure 2: The three sequential experimental phases: Existence, Mitigation, and Utilization.

Key Empirical Findings

- **Single-Answer vs. Pairwise Gap:** Absolute grading forces judges to evaluate in isolation, leading to high internal doubt. Pairwise comparisons provide relative anchors, resulting in highly decisive and confident assessments.
- **Self-Preference Bias:** Evaluations within the same model family (e.g., Llama assessing Llama) result in artificially inflated ratings paired with unwarranted confidence spikes due to shared pre-training and stylistic overlap.
- **The Capability Paradox:** General capability on reasoning benchmarks does not prevent evaluation doubt. Highly capable models like GPT-4o exhibit high uncertainty in default grading formats compared to less advanced models.

ConfiLM OOD Performance

- **OOD Challenges:** Standard judges fail in highly specific OOD domains (like real-time sports results) because they rely on in-distribution training data and overlook factual inaccuracies.
- **The "Rugby Sevens" Case Study:** When a candidate model generated a news report with a fabricated date, GPT-4o and standard fine-tuned models chose the hallucinated, verbose response.
- **The ConfiLM Victory:** By ingesting the verbalized low-confidence of the candidate, ConfiLM correctly flagged the response as inaccurate and penalized the hallucination.

Core Framework Strengths

- **Dual-Uncertainty Interface:** Establishes the first unified framework integrating candidate generation doubt with judge decision confidence.
- **Zero-Cost Calibration:** Demonstrates that Chain-of-Thought (CoT) acts as a computationally free, inference-time logit stabilizer.
- **The Verbalization Bridge:** Successfully bypasses the numerical limits of language models by translating float values into natural adjectives.
- **Expert-Curated Benchmark:** The Olympic 2024 dataset provides a rigorous, highly aligned OOD testbed for automated judges.

Operational Limitations

- **Closed-Source API Barrier:** Logit extraction is impossible on proprietary commercial platforms (such as Claude and Gemini) that withhold token probabilities.
- **Inference Latency Overhead:** Forcing a detailed explanation before outputting the score token elevates generation costs and runtime latency.
- **Exposure Bias in SFT:** Specialized judges fine-tuned on rigid templates suffer from high scoring volatility when exposed to unfamiliar domains.

Critical Strategic Outlook

- **Confidence-Triggered Escalation:** Future AI-evaluation platforms should establish automated confidence thresholds, triggering selective escalation to human raters for low-confidence judgments.
- **Multimodal Uncertainty:** Extending the framework to measure uncertainty across image, audio, and video inputs in multi-modal judges.
- **Multi-Turn Conversation Decay:** Investigating how evaluation uncertainty propagates or decays across complex, multi-turn dialogues.

Paper 4:
Preference Leakage: A CONTAMINATION PROBLEM IN LLM-AS-A-JUDGE

Introduction & Category Alignment

Context: The Evolution of Model Evaluation

- **Shift in Evaluation Paradigms**:
  - Traditional lexical n-gram metrics (BLEU, ROUGE) fail to capture semantic nuances in open-ended LLM outputs.
  - **LLM-as-a-Judge** has emerged as the industry standard for scalable, fast, and human-aligned model benchmarking.
- **The Synthetic Data Engine**:
  - Modern LLM alignment relies heavily on synthetic datasets generated by state-of-the-art teacher models (e.g., GPT-4o).
- **The Core Vulnerability**:
  - When the model producing the training data and the model judging the student share architectural or family ties, systemic evaluation bias occurs.

Category Alignment: Data vs. Preference Leakage

- **Data Leakage (Classical Contamination)**:
  - Direct overlap between pre-training/fine-tuning datasets and evaluation benchmark test sets.
  - Model simply memorizes specific ground-truth questions and answers.
- **Preference Leakage (Evaluator Contamination)**:
  - Structural, familial, or distillation relatedness between Data Generator (M_G) and Judge (M_J).
  - Preferences are transferred to Student (M_S) via synthetic training data (D_{syn}).
  - Evaluator inflates scores based on inherited stylistic and structural priors rather than objective response correctness.

Overview Diagram: Preference vs. Data Leakage

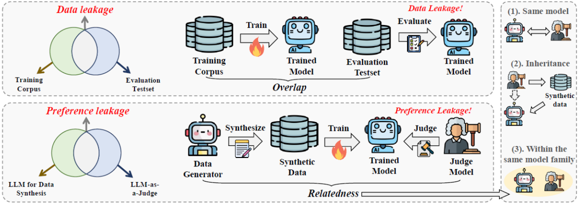

Figure 1: Overview of preference leakage. We make a comparison between data leakage and preference leakage and present three types of relatedness: being the same model, having an inheritance relationship and belonging to the same model family.

Figure 3: Conceptual Architecture of Preference Leakage and Model Lineage Relationships

Methodology & Experimental Pipeline

End-to-End Experimental Pipeline

- **Stage 1: Seed Sampling & Prompt Extraction**:
  - Sampled 30,000 diverse open-ended prompts across diverse domains from UltraFeedback.
- **Stage 2: Synthetic Data Generation ($M_G \rightarrow D_{syn}$)**:
  - Teacher models (M_G) generate high-quality candidate responses to form instruction-tuning datasets.
- **Stage 3: Student Model Post-Training (M_S)**:
  - Base models (Mistral-7B, Qwen-2.5-14B) are trained on D_{syn} via SFT, DPO, or In-Context Learning (ICL).
- **Stage 4: Pairwise Evaluation & PLS Calculation**:
  - Judge models (M_J) evaluate M_S against baseline models on Arena-Hard and AlpacaEval 2.0 to quantify score inflation.

Part 1: Problem Definition & Lineage Taxonomy

- **Conceptual Formulation**:
  - Evaluation score assigned by M_J to M_S is systematically inflated when M_G is related to M_J.
- **Three Lineage Relatedness Categories**:
  1. **Same Model**: $M_G \equiv M_J$ (e.g., GPT-4o generates synthetic training data and later evaluates the student model).
  2. **Inheritance**: M_J is fine-tuned directly on M_G's outputs or shares an identical fine-tuning lineage.
  3. **Same Model Family**: M_G and M_J share architectural design, tokenizers, and pre-training data corpora (e.g., GPT-4o and GPT-4-Turbo).
- **Key Novelty #1**:
  - First formal taxonomy defining evaluator-generator dependency as an implicit bias channel distinct from simple self-enhancement.

Part 2: Experimental Controls & Metric Mechanics

- **Controlled Experimental Setup**:
  - **Data Generators / Judges**: GPT-4o, Gemini-1.5-Flash, LLaMA-3.3-70B.
  - **Base Students**: Unaligned base models to eliminate pre-existing instruction-tuning contamination.
  - **Benchmarks**: Arena-Hard (500 challenging prompts) and AlpacaEval 2.0 (805 prompts).

- **Preference Leakage Score (PLS) Formulation**:
  - $PLS = \text{WinRate}(M_S \text{ judged by Related } M_J) - \text{WinRate}(M_S \text{ judged by Control } M_{ctrl})$.
  - Controls for position bias (swapping model order) and length bias.

- **Key Novelty #2**:
  - A rigorous cross-evaluation benchmarking framework that isolates style inheritance from genuine reasoning quality.

Part 3: Mechanistic Probing & Stealth Bias

- **The Recognition Paradox**:
  - Can judges consciously detect outputs from their related student models?
  - Direct probing reveals LLM judges perform near random chance (30% – 53% accuracy) at identification.
- **Classifier Detectability**:
  - A simple linear classifier or lightweight BERT model achieves over 82.4% accuracy in identifying student generations.
- **Spurious Features as Conduits**:
  - Bias is driven by surface-level artifacts: formatting templates, markdown headers, bullet-point frequency, and tone.
- **Key Novelty #3**:
  - Proving preference leakage is a subconscious phenomenon where LLM judges reward stylistic alignment without explicit awareness.

Key Empirical Results (Verbal Summary)

- **Widespread Score Inflation**:
  - Significant positive PLS across all major model families (reaching up to 37.1% win-rate inflation on Arena-Hard).
- **Model Scale Impact**:
  - Smaller student models (1B–3B parameters) show **higher** leakage scores because they memorize surface-level response templates more aggressively.
- **Post-Training Method Dynamics**:
  - **SFT**: Maximum leakage (23.6% average) due to heavy stylistic imitation.
  - **DPO**: Significantly reduced leakage (5.2%) by penalizing stylistic repetition.
  - **ICL**: Minimal to zero leakage (-2.7%) as core weights remain untouched.
- **Domain & Task Sensitivity**:
  - Highest bias in open-ended creative writing and coding; lowest in deterministic math and logic problems.

Strengths & Methodological Rigor

- **Pioneering Problem Formulation**:
  - Uncovers a systemic vulnerability in the synthetic data lifecycle that undermines community leaderboard integrity.
- **Methodological Isolation**:
  - Isolates confounding variables by starting exclusively with raw base models, avoiding instruction-tuning artifacts.
- **Broad Practical Impact**:
  - Demonstrates real-world implications for major public evaluations such as LMArena, AlpacaEval, and Open LLM Leaderboard.

Limitations & Experimental Constraints

- **High Computational & API Costs**:
  - Full pairwise evaluation matrices across frontier models demand massive API query volume and compute overhead.
- **Proprietary Black-Box Constraints**:
  - Closed-source model providers do not disclose full pre-training data or exact model lineages, limiting complete tracking.
- **Inefficacy of Simple Prompting Mitigation**:
  - Standard system prompts or Chain-of-Thought (CoT) instructions fail to remove preference leakage without structural adjustments.

Future Directions & Mitigation Strategies

- **Strict Lineage Isolation**:
  - Public leaderboards must enforce strict separation between data generation teacher models and benchmark judge models.
- **Post-Hoc Statistical Calibration**:
  - Use held-out human judgment samples to dynamically adjust and normalize judge scoring distributions.
- **Multi-Judge Ensembling**:
  - Combine evaluations from multiple non-related model families to cancel out model-family bias.
- **Stylistic Normalization Pipelines**:
  - Pre-process responses through neutralizing paraphrasers to strip surface-level formatting artifacts before judging.

Summary & Key Takeaways

- **Takeaway 1**: Preference leakage is an insidious, subconscious contamination mechanism in modern LLM evaluation.
- **Takeaway 2**: Judges favor student models that mirror their surface-level style, formatting, and structural cadence.
- **Takeaway 3**: SFT intensifies leakage, whereas DPO, ICL, and post-hoc calibration offer effective defense pathways.
- **Core Recommendation**: Decouple evaluation judges completely from synthetic training data pipelines for trustworthy benchmarks.